sense of those observations. Only in the chapter after that will we offer a semi-historical account of the discoveries that led people to contemplate such a dramatically new kind of physics in the first place. Then we'll hammer home exactly how dramatic some of the implications of quantum mechanics really are.

With all that in place, over the rest of the book we can set about the fun task of seeing where all this leads, demystifying the most striking features of quantum reality.

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Physics is one of the most basic sciences, indeed one of the most basic human endeavors. We look around the world, we see it is full of stuff. What is that stuff, and how does it behave?

These are questions that have been asked ever since people started asking questions. In ancient Greece, physics was thought of as the general study of change and motion, of both living and nonliving matter. Aristotle spoke a vocabulary of tendencies, purposes, and causes. How an entity moves and changes can be explained by reference to its inner nature and to external powers acting upon it. Typical objects, for example, might by nature be at rest; in order for them to move, it is necessary that something be causing that motion.

All of this changed thanks to a clever chap named Isaac Newton. In 1687 he published *Principia Mathematica*, the most important work in the history of physics. It was there that he laid the groundwork for what we now call "classical" or simply "Newtonian" mechanics. Newton blew away any dusty talk of natures and purposes, revealing what lay underneath: a crisp, rigorous mathematical formalism with which teachers continue to torment students to this very day.

Whatever memory you may have of high-school or college homework assignments dealing with pendulums and inclined planes, the basic ideas of classical mechanics are pretty simple. Consider an object such as a rock. Ignore everything about the rock that a geologist might consider interesting, such as its color and composition. Put aside the possibility that the basic structure of the rock might change, for example, if you smashed it to pieces with a hammer. Reduce your mental image of the rock down to its most